

Vibe Decoding*

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Abstract

Greedy decoding in large language models can generate globally incoherent sequences that fail to capture broader semantic patterns critical for success on certain tasks. Existing alternatives to greedy decoding that approximate exact search or introduce inductive biases can be computationally expensive or require running an auxiliary model alongside the main model. In this paper, we introduce *vibe decoding*, a novel, computationally efficient, and auxiliary-model-free decoding method that aims to give a model semantic foresight by preferring next tokens which align with the expected semantic direction of the model in embedding space. Then, we characterise the asymptotic runtime of the method, and the hyperparameter interval for which the method behaves identically to greedy decoding. Finally, we study how the method can outperform and underperform greedy decoding.

1 Introduction

Large language models (LLMs) trained on massive textual corpora have shown an ability to perform tasks from textual instructions (Brown et al., 2020). LLMs often decode autoregressively, growing token sequences stepwise by repeatedly generating and appending the most probable next token at each step. This method of decoding, known as *greedy decoding*, is the de facto method of approximating maximum a posteriori (MAP) sequences, as exact solutions are intractable in theory. Greedy decoding can generate globally incoherent sequences that fail to capture broader semantic patterns critical for success on certain tasks (Garrido-Munoz et al., 2025). This limitation has motivated a growing body of work exploring alternative decoding methods that approximate exact search or introduce

inductive biases. However, these methods can be computationally expensive or require running an auxiliary model alongside the main model.

In this paper, we introduce *vibe decoding*, a novel, computationally efficient, and auxiliary-model-free decoding method that aims to give a model semantic foresight by preferring next tokens which align with the expected semantic direction of the model in embedding space.

2 Existing Work

For better performance on various tasks, various decoding methods have been proposed. A few have emerged as broadly applicable methods that are now widely adopted in production systems.

Lookahead methods approximate exact search by computing probabilities of prospective continuations before committing to a next token. In particular, beam search (Lowerre, 1976) uses breadth-first search (BFS) with top- B pruning by log-probability. This aims to retain less probable token sequences that could become more probable after growth. However, this can be computationally expensive, as there are often many tokens to consider at each step.

Contrastive methods introduce an inductive bias for continuations that, between an expert model and an amateur model, are uniquely preferred by the expert model. In particular, contrastive decoding (Li et al., 2023) uses the decoding rule

$$\arg \max_{y \in V} \left[\log \Pr_{\Delta}(y | \mathbf{x}) - \log \Pr_{\nabla}(y | \mathbf{x}) \right] \quad (1)$$

where $\Pr_{\Delta}$ is the expert model, and $\Pr_{\nabla}$ is the amateur model. This aims to factor out undesired behaviours highlighted by smaller, amateur language models from the good behaviours of larger, expert language models. However, this requires running an auxiliary model alongside the main model.

*Undergraduate final project for 50.040 Natural Language Processing at the Singapore University of Technology and Design. Code repository at github.com/gregorylimeurhen/vibe-decoding.

3 Methodology

We define the vibe decoder as follows.

Let V be a finite vocabulary, and $d \in \mathbb{N}_{\geq 1}$ be a hidden size, and $\mathbf{W} \in \mathbb{R}^{|V| \times d}$ be an embedding matrix.

First, we define

$$\mathbf{E} = \left[\frac{\mathbf{w}_i}{\|\mathbf{w}_i\|} \right]_{i=1}^{|V|}$$

as the **vibe matrix**. This unit normalisation retains directional information and discards magnitude information.

Then, we define

$$\mathbf{d}_x = \mathbb{E}_{c \sim \Pr(\cdot | x)} [\mathbf{e}_c]$$

as the **vibe direction**, where $x \in V^*$ is an input. This can be seen as the expected semantic direction of the model in embedding space.

Finally, we define

$$f_\lambda : V^* \rightarrow V$$

$$f_\lambda(x) \in \arg \max_{y \in V} [\log \Pr(y | x) + \lambda \langle \mathbf{e}_y, \mathbf{d}_x \rangle]$$

as the **vibe decoder**, where $\lambda \in \mathbb{R}$ is a vibe strength. This augments the greedy decoding rule with a vibe alignment term that favours tokens which align with the vibe direction. Notably, f_0 is the greedy decoder.

We present two theorems concerning the vibe decoder as follows.

Let $\mathcal{L}(y, x) = \log \Pr(y | x)$ be the likelihood term, and $\mathcal{V}(y, x) = \langle \mathbf{e}_y, \mathbf{d}_x \rangle$ be the vibe alignment term.

Theorem 1 (Time Complexity of Vibe Decoding). Let $\mathcal{T}(\varphi)$ denote the time complexity of evaluating a function φ . Then,

$$\mathcal{T}(f_\lambda) \in \Theta(|V|d)$$

for each vibe strength $\lambda \in \mathbb{R}$.

This theorem characterises the asymptotic runtime of vibe decoding, and shows the runtime to be identical to that of greedy decoding. We give a proof of Theorem 1 in Appendix B.

Theorem 2 (Inactive Interval of Vibe Decoding).

Let $x \in V^*$ be input, and $g = f_0(x)$ be the token selected by the greedy decoder, and

$$\wedge_{x,g}(\lambda) = \max_{y: \mathcal{V}(y,x) < \mathcal{V}(g,x)} \frac{\mathcal{L}(g,x) - \mathcal{L}(y,x)}{\mathcal{V}(y,x) - \mathcal{V}(g,x)}$$

be the lower inert threshold on vibe strength, and

$$\vee_{x,g}(\lambda) = \min_{y: \mathcal{V}(y,x) > \mathcal{V}(g,x)} \frac{\mathcal{L}(g,x) - \mathcal{L}(y,x)}{\mathcal{V}(y,x) - \mathcal{V}(g,x)}$$

be the upper inert threshold on vibe strength. Then,

$$g = f_\lambda(x) \iff \lambda \in [\wedge_{x,g}(\lambda), \vee_{x,g}(\lambda)]$$

for each vibe strength $\lambda \in \mathbb{R}$.

This theorem characterises the interval of vibe strengths for which the method behaves identically to greedy decoding.

4 Experimentation

We experimented with three models: Qwen 2.5 0.5B (Instruct), Gemma 2 2B (Instruct), and Llama 3.1 8B (Instruct).

Model	Parameter count
Qwen 2.5 0.5B	494,032,768
Gemma 2 2B	2,614,222,080
Llama 3.1 8B	8,030,261,248

Table 2: Model parameter counts.

Our experiments include four datasets: BoolQ, NumerSense, PG19, and WikiText-2. For each dataset, we adopted a task and metric suggested by the Hugging Face community.

Dataset	Task
BoolQ	Text classification
NumerSense	Mask filling
PG19	Language modeling
WikiText-2	Language modeling

Table 3: Table showing datasets and their tasks.

Our experiments adopt three metrics: macro F1, mean reciprocal Rank (MRR), and perplexity (PPL).

Dataset	Length	Split	Prefix
BoolQ	16k rows	validation	All
NumerSense	3k rows	validation	All
PG19	11B chars	test	10
WikiText-2	13M chars	validation	1,000

Table 4: Table showing dataset restrictions.

Dataset	Metric	Model	$\lambda = -5$	$\lambda = 0$	$\lambda = 1$	$\lambda = 5$
BoolQ	Macro F1	Qwen 2.5 0.5B	0.4091	0.5393	0.5623	0.6396
		Gemma 2 2B	0.7701	0.7879	0.7923	0.7980
		Llama 3.1 8B	0.7008	0.7471	0.7570	0.7902
NumerSense	MRR	Qwen 2.5 0.5B	0.3839	0.4117	0.4160	0.4309
		Gemma 2 2B	0.4634	0.4709	0.4721	0.4788
		Llama 3.1 8B	0.5380	0.5996	0.6117	0.6508
PG19	PPL	Qwen 2.5 0.5B	40.4268	27.1570	27.6489	34.7165
		Gemma 2 2B	35.7951	23.0967	23.3504	29.9163
		Llama 3.1 8B	31.6030	18.4746	19.0450	25.6222
WikiText-2	PPL	Qwen 2.5 0.5B	13.8632	9.1044	9.2280	11.1649
		Gemma 2 2B	24.7163	16.3177	17.1099	23.7915
		Llama 3.1 8B	8.2376	5.3852	5.4349	6.4275

Table 1: Table showing performance of models on BoolQ, NumerSense, PG19, and WikiText-2 across values of λ .

BoolQ (Clark et al., 2019) contains (q, a) pairs, where q is a question, and a is either “true” or “false”. According to the authors, the questions are naturally occurring, as they are generated in unprompted and unconstrained settings.

NumerSense (Lin et al., 2020) contains (s, t) pairs, where s is a sentence with one numeric token masked, and $t \in \mathbb{N}_{[0,10]}$ is a target. According to the authors, finding the correct numeric token to fill the masked slot in each sentence involves numerical common sense.

PG19 (Rae et al., 2019) contains text from books in Project Gutenberg. According to the authors, the corpus is designed as a long-range language modelling benchmark, with documents that are substantially longer than those in prior datasets.

WikiText-2 (Merity et al., 2016) contains text from Good and Featured articles on Wikipedia. According to the authors, the questions are naturally occurring, as they are generated in unprompted and unconstrained settings.

We experimented with vibe strengths $-5, 0, 1, 5$. Hereafter, we let λ denote a vibe strength.

Our code repository at

[github.com/gregorylimeurhen/
vibe-decoding](https://github.com/gregorylimeurhen/vibe-decoding)

contains our exact experimental configurations. Due to resource constraints, our experiments were limited to graphics processing units (GPUs) available through short-term cloud credits.

We present two positive results and two negative

results as follows.

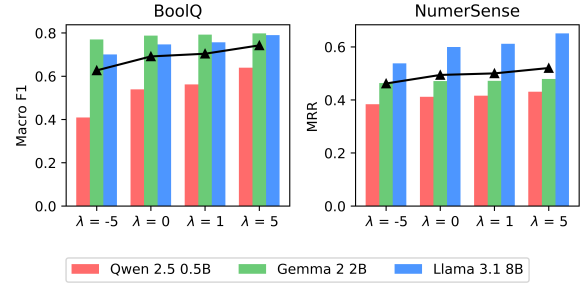


Figure 1: Plot showing performance of models on BoolQ and NumerSense across values of λ .

On the BoolQ and NumerSense tasks, increasing λ from -5 to 5 consistently improved Macro F1 and MRR across all models, with $\lambda = 5$ yielding the best result.

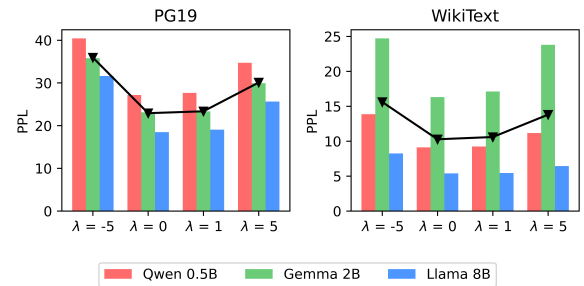


Figure 2: Plot showing performance of models on PG19 and WikiText-2 across values of λ .

On the PG19 and WikiText-2 tasks, changing λ from 0 to any other value consistently degraded PPL across all models.

Overall, these positive and negative results seem to

suggest the following.

1. Vibe decoding could outperform greedy decoding on tasks that benefit from answering questions involving general world knowledge and numerical commonsense reasoning.
2. Vibe decoding could underperform greedy decoding on tasks that require strict training distribution adherence.

On the other tasks we experimented with, the results for vibe decoding and greedy decoding were extremely close (and often tied).¹ This phenomenon can be explained by embedding space isotropy.

Let $\mathbf{x} \in V^*$, and $g = f_0(\mathbf{x})$ be the token selected by greedy decoding, and

$$\epsilon_{\mathbf{x}} = \max_{y \in V} |\mathcal{V}(y, \mathbf{x}) - \mathcal{V}(g, \mathbf{x})|$$

be the vibe dispersion of $\mathbf{x}$, and

$$\delta_{\mathbf{x}} = \min_{y \neq g} (\mathcal{L}(g, \mathbf{x}) - \mathcal{L}(y, \mathbf{x}))$$

be the log-probability margin of g .

Under an idealised isotropic model, the projections in

$$\{\mathcal{V}(y, \mathbf{x}) : y \in V\}$$

concentrate tightly around 0, so that

$$\epsilon_{\mathbf{x}} \in O\left(\|\mathbf{d}_{\mathbf{x}}\| \sqrt{\frac{\log|V|}{d}}\right)$$

with high probability.

If $\delta_{\mathbf{x}} > 0$, then for each λ satisfying $|\lambda| \leq \frac{\delta_{\mathbf{x}}}{\epsilon_{\mathbf{x}}}$, the perturbation

$$\lambda \mathcal{V}(y, \mathbf{x})$$

is too small to change the selected token.

We also studied the embedding space isotropy of each model empirically as follows.

First, we define

$$r(\varepsilon) = \frac{|\{(i, j) \in V^2 : |\langle \mathbf{e}_i, \mathbf{e}_j \rangle| \leq \varepsilon\}|}{|V|^2}$$

as the embedding ε -isotropy profile, where $\varepsilon \in \mathbb{R}_{[0,1]}$. This can be seen as the fraction of token pairs with similarity no greater than ε .

¹We give a list of the datasets for these tasks in Appendix A.

Then, we use Monte Carlo sampling to approximately plot $r(\varepsilon)$ against ε for each model.

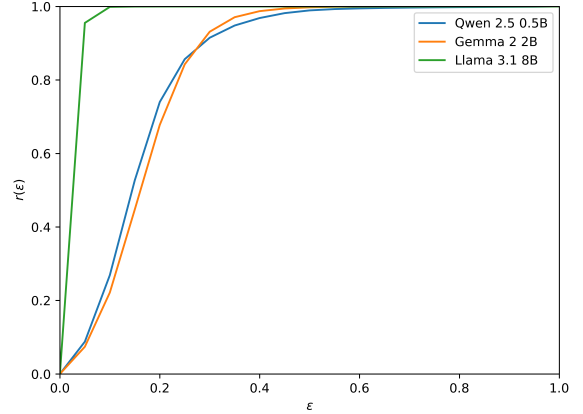


Figure 3: Plot showing $r(\varepsilon)$ against ε .

Finally, we computed the area under the curve (AUC)

$$\int_0^1 r(\varepsilon) d\varepsilon$$

for each plot.

Model	$r(\varepsilon)$ - ε AUC
Qwen 2.5 0.5B	0.8378
Gemma 2 2B	0.8322
Llama 3.1 8B	0.9727

Table 5: Table showing $r(\varepsilon)$ - ε AUC of models.

Interestingly, across all models, the AUC is close to 1. This suggests that each model has a high degree of embedding space isotropy, as a high AUC implies many token pairs have low similarity.

5 Conclusion

In this paper, we introduced *vibe decoding*, a novel, computationally efficient, and auxiliary-model-free decoding method that aims to give a model semantic foresight by preferring next tokens which align with the expected semantic direction of the model in embedding space.

Then, we characterised the asymptotic runtime of the method, and the hyperparameter interval for which the method behaves identically to greedy decoding.

Finally, we studied how the method can outperform and underperform greedy decoding.

6 Future Work

Our work admits many possible extensions. For example, one could investigate the performance of

vibe decoding on larger models, explore adaptive mechanisms for tuning vibe strength based on context, or combine vibe decoding with other decoding methods.

7 Acknowledgements

This work was completed as part of an undergraduate final project for 50.040 Natural Language Processing at the Singapore University of Technology and Design.

We thank the teaching staff for their guidance throughout the course, and appreciate the feedback and support provided throughout the project.

8 Ethics Statement

This work only used publicly available datasets and models. GPU usage was minimised.

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A Datasets

The unpresented datasets we explored include CNN / DailyMail, ARC (AI2), RealToxicityPrompts, CommitmentBank, COPA, PIQA, AQUA-RAT, LAMBADA, WSC, PTB, AG News, Yelp Review Polarity, Poem Sentiment, HotpotQA, CoLA, WNLI, GSM8K, HumanEval, PubMedQA, SQuAD, TinyStories, HellaSwag, and SST-2.

B Proofs

Proof of Theorem 1. Greedy decoding requires two steps: computing log-probabilities over V , and selecting the token with maximal probability. Computing the log-probabilities requires $\Theta(|V|d)$ time. Selecting the token requires $\Theta(|V|)$ time. Thus, greedy decoding requires

$$\Theta(|V|d) + \Theta(|V|) = \Theta(|V|d)$$

time. Vibe decoding adds exactly two steps to greedy decoding: computing the vibe direction, and computing the alignment term for each candidate token. Each step requires $\Theta(|V|d)$ time. Thus, vibe decoding requires

$$\begin{aligned} \Theta(|V|d) + \Theta(|V|d) + \Theta(|V|d) &= 3\Theta(|V|d) \\ &= \Theta(|V|d) \end{aligned}$$

time. □